Lecture 15

FAIR in Practice

Interoperability · Reusability · Reproducibility

🌱 **Guest contribution: Jamie McCann introduced PlantGenIE live in Lecture 14**

In these exercises you will query real research infrastructure from the command line — demonstrating FAIR in your own institution's primary research data.

Session Aims & ILO Links

1	Query a REST API from the command line and explain how APIs embody the Interoperable principle	ILO 1, 3
2	Explain PlantGenIE as an example of research infrastructure designed for interoperability	ILO 3
3	Apply a metadata sufficiency checklist to a real GEO/ENA dataset	ILO 3, 13
4	Reconstruct a bioinformatics analysis pipeline from a methods section and identify reproducibility gaps	ILO 3, 13
5	Commit all exercise outputs, commands, and findings to Git with structured documentation	ILO 12
6	Apply the FAIR essay marking criteria to your own preliminary paper assessment	ILO 13

These are self-study exercises. Bring questions from Lecture 14 to office hours or the Canvas discussion board. The essay assessment closes [date — see Canvas].

FAIR Applied: Exercise Overview

Each exercise maps directly to a FAIR principle — and to the essay assessment.

I — Interoperable

🌱 PlantGenIE API

Query the PlantGenIE REST API from the command line. Retrieve gene information, expression data, and BLAST results in standard formats. Understand why API design choices either enable or limit FAIR compliance.

Builds on Jamie McCann's Lecture 14 introduction

R — Reusable

📊 GEO/ENA Metadata Assessment

Apply the metadata sufficiency checklist to a real dataset. Determine what is present, what is missing, and what the practical consequences of each gap would be for a researcher attempting reanalysis.

Independent exercise

R — Reusable

🔬 Paper Reproducibility Exercise

Reconstruct the analysis pipeline from your chosen essay paper. Build the evidence base you will use in the essay. Identify the most significant reproducibility gaps.

Independent exercise using your essay paper

Part 1

I = Interoperable: The PlantGenIE API

PlantGenIE: Research Infrastructure at Umeå

🌱 *Developed at Umeå University · plantgenie.se · github.com/plantgenie*

🌲 Genome browsers

Reference genomes for Picea abies (Norway spruce) and Populus tremula (European aspen), with gene models, annotations, and synteny tools

📈 Expression data

Integrated RNA-seq expression atlases across tissues and conditions, with co-expression network tools for identifying gene modules

🔍 BLAST and search

Sequence similarity search against conifer and aspen genomes — accessible via web interface and API

⚡ REST API

All data accessible programmatically via documented HTTP endpoints returning JSON and FASTA. You will use this throughout these exercises.

Jamie McCann introduced PlantGenIE and its API design decisions live in Lecture 14 — the exercises below put that into practice.

Why APIs Embody the Interoperable Principle

A well-designed API means: any tool, any programming language, any researcher — access the same data in the same way.

I1 **Uses a formal, accessible, broadly applicable language**

REST over HTTPS — the same protocol every web browser uses. Any tool that can make HTTP requests can query the API.

I2 **Uses vocabularies that follow FAIR principles**

Gene entries include GO term identifiers (GO:XXXXXXX), InterPro accessions, NCBI taxonomy IDs — stable, machine-readable ontology terms, not free text.

I3 **Includes qualified references to other data**

Gene entries cross-reference ENA accessions, Ensembl gene IDs, UniProt accessions — forming a web of connected resources.

+ **Documented endpoints**

The API documentation describes what each endpoint accepts and returns. A researcher can query it without contacting the authors or using a specific browser.

Compare: a database that only provides data via a browser download, with no stable URL and no documented structure, is not Interoperable even if the data is technically available.

Exercise 1: Querying the PlantGenIE API

Login to Kebnekaise via OnDemand. Full commands in your handout.

📁 Setup	Create lecture15-fair-practice directory · git init · initial commit
🧬 1A — Gene info	Query gene information for a Populus tremula gene (POST request) · inspect JSON response · extract GO terms and cross-references
📊 1B — Expression	Retrieve expression data for the AspWood experiment · identify the most highly expressed sample · note whether expression units are specified
🔍 BLAST via PlantGenIE	PlantGenIE's BLAST endpoint is still being finalised — demonstrated live by Jamie McCann. The three ways to run BLAST (1C–1F) follow on the next slides.
📝 Commit your progress	Document 1A/1B findings in README · git add · git commit — you'll add BLAST findings and a final summary commit in 1F

If plantgenie.se is unavailable, check the current API documentation or post on the Canvas discussion board.

Exercise 1: Key API Commands

Parts 1A and 1B — gene information and expression

```
$ BASE="https://www.plantgenie.se/api"
$ SPECIES="populus-tremula"; GENE_ID="Potra2n4c9093"
# 1A: Retrieve gene info (POST with JSON body)
$ curl -s -X POST "${BASE}/v1/annotations" \
$   -d "{\"species\":\"${SPECIES}\",\"geneIds\":[\"${GENE_ID}\"]}" > gene_info.json
# 1B: Retrieve expression data (AspWood experiment, ERP016242)
$ curl -s -X POST "${BASE}/v1/expression" \
$   -d "{\"experimentId\":15,\"geneIds\":[\"${GENE_ID}\"]}" > expression_data.json
$ cat gene_info.json expression_data.json
```

Part 1C — BLAST via PlantGenIE (not yet finalised)

```
# PlantGenIE also exposes a BLAST endpoint.
# It will be demonstrated live once the new API's
# endpoint and request format are confirmed.

# For a working BLAST API today, see Exercise 1E:
# NCBI's BLAST API (submit → poll → retrieve)



$ git add . && git commit -m "lecture15: PlantGenIE API exercises complete"
```

Exercise 1 continued — Three Ways to Run BLAST

Same TP53 sequence, three tools, one workflow. Full commands in your handout.

🌐 1C — Web BLAST	Run blastp via the NCBI or PlantGenIE website - note the parameters used and how long the search takes
💻 1D — Local HPC	Submit the same search as a Slurm job on Kebnekaise against the local Swiss-Prot database - compare setup effort to the website
🔌 1E — API	Submit via the NCBI BLAST API (submit, poll, retrieve) - a working, documented example of the submit/poll pattern PlantGenIE's own BLAST endpoint will likely also use once finalised
📊 1F — Compare and Commit	Complete the comparison table (speed, scalability, reproducibility, control) in your README and commit your findings
💡 Reflect	Which approach would you choose for 1 sequence versus 500? Why does that change depending on whether reproducibility matters?

Background on how BLAST works (seeding, E-values, scoring) is in the Lecture 12 reference material - read it before starting this part if you have not already.

Part 2

R = Reusable: Metadata assessment and paper reproducibility

The Reusability Challenge: Can You Actually Reuse This Data?

Deposited data is not the same as reusable data.

Scenario: not reusable

A GEO entry has 12 RNA-seq samples. Raw FASTQ files are available. The samples are labelled "sample_01" through "sample_12". The GEO description says "expression profiling of treated vs untreated cells." There are no ontology terms, no specification of cell type, no indication of which samples are treated and which are control, and the reference genome version is not stated.

Result: a researcher who finds this dataset cannot determine experimental conditions without contacting the authors. The data exists, but it is not reusable.

Scenario: reusable

The same 12 samples, but each is annotated with: organism (NCBI:txid9606), tissue (UBERON:0002368), treatment (EFO:0001461 for dexamethasone treatment), concentration and duration, and a link to the reference genome accession (GCF_000001405.40). Each condition has 3 biological replicates clearly labelled. Analysis code is available with a Zenodo DOI.

Result: an independent researcher can immediately reproduce the differential expression analysis from the raw data alone.

The difference is not the data — it is the metadata. This is exactly what your exercise assesses.

Exercise 2: The Metadata Sufficiency Checklist

Dataset: ERP016242 · Sundell et al. (2017) AspWood · Populus tremula wood formation · The Plant Cell

Sample identification

- ☐ Findable by search without knowing accession in advance?
- ☐ All samples identifiable and assignable to experimental conditions?
- ☐ Biological replicates clearly labelled?

Biological metadata

- ☐ Organism with NCBI taxonomy ID?
- ☐ Tissue/cell type with UBERON or CL ontology term?
- ☐ Treatment with EFO or CHEBI ontology term?
- ☐ Other variables (age, sex, genotype, time point)?

Technical metadata

- ☐ Sequencing platform specified?
- ☐ Library type (stranded, poly-A/ribo-depl.) specified?

Exercise 3: Paper Reproducibility Assessment

Open your chosen essay paper. Work through the following systematically.

1. **Map the analysis pipeline**

   List every computational step in the methods. For each: tool, version specified?, parameters reported? Build a table — see example below.

2. **Assess code availability**

   Is code available at all? If yes: is it versioned (Git tag/Zenodo), documented, with dependencies pinned? If no: is there any reproducibility path?

3. **Assess the reference genome**

   Is the reference genome specified? Is the assembly accession version given? Is the annotation GTF version specified? Could you identify exactly the right reference to reproduce the alignment?

4. **Write a reproducibility verdict**

   2-3 sentences: could an independent researcher reproduce the main analysis? What specific information would they need from the authors that is not in the paper?

Example pipeline table in your handout. This becomes the evidence base for your essay — take careful notes.

Example: Reconstructing the Analysis Pipeline

Build this table from the methods section of your chosen paper. This is your FAIR evidence base.

Step	Tool	Version?	Parameters?	Notes
Quality trimming	Trimmomatic	v0.39 ✓	LEADING:3 TRAILING:3 MINLEN:36 ✓	Well reported
Alignment	HISAT2	v2.1.0 ✓	'default parameters' ✗	Parameters missing
Quantification	featureCounts	Not specified ✗	Not specified ✗	Version + params missing
DE analysis	DESeq2	v1.20.0 ✓	Shrinkage method not stated ✗	Partial — key parameter absent
Reference genome	GRCh38	Release not specified ✗	—	Cannot confirm exact version

'Default parameters' is almost never sufficient for reproducibility — defaults change between software versions, and many tools have multiple default modes depending on other settings. This is one of the most common reproducibility failures in bioinformatics methods sections.

Part 4 — Building a Reproducible Report

Turn a small piece of this exercise into a rerunnable notebook. Full commands in your handout.

📓 Build the notebook	Launch Jupyter via OnDemand · create 3 cells: a markdown description, code that loads and summarises expression_data.json, and a markdown interpretation
🔁 Restart & Run All	Kernel → Restart & Run All — confirms the notebook reproduces its own output from scratch, not just that it ran once while you were writing it
💾 Export & commit	Export to HTML (File → Download as → HTML) · git add · commit both the notebook and its HTML export
📄 The R equivalent	Same idea in R uses R Markdown + knitr via RStudio Server — you'll use this properly in Data Science for Biology with R. Not required for this exercise.
💡 Reflect	What did the notebook make explicit that a bare script or terminal history would not? Could a labmate reproduce your output from the .ipynb file alone?

See Lecture 14, Section 4.5 for why notebooks and R Markdown matter for reproducibility.

Essay Assessment

From these exercises to the Canvas assignment — full assessment details in Lecture 14, Section 10

From Exercise to Essay: The Direct Connection

Everything you do in these exercises is direct preparation for the essay.

Exercise: PlantGenIE API exercise

→ Essay: Evidence for the Interoperable principle — you have now experienced what a well-designed API enables. Use this as a reference point when assessing I in your essay paper.

Exercise: GEO metadata checklist

→ Essay: Direct template for your Reusability assessment. Apply the same checklist to your essay paper's dataset. Your completed checklist is your methodology.

Exercise: Pipeline reconstruction table

→ Essay: This IS your reproducibility evidence. The table you build here becomes the backbone of your essay's computational reproducibility section.

Exercise: Reproducibility verdict

→ Essay: The 2-3 sentence verdict you write here is the draft of your essay's synthesis paragraph. Develop it with specific consequences and recommendations.

What You Should Know After These Exercises

- ✓ Query a REST API from the command line and parse JSON responses with grep
- ✓ Explain how APIs embody FAIR Interoperability — standard protocols, ontology terms, cross-references, documented endpoints
- ✓ Describe PlantGenIE as research infrastructure designed for programmatic access to boreal tree genomics data
- ✓ Apply the metadata sufficiency checklist to a GEO/ENA dataset and identify specific reusability gaps
- ✓ Reconstruct a bioinformatics analysis pipeline from a methods section, noting version and parameter gaps
- ✓ Write a reproducibility verdict identifying what is missing and what it would prevent
- ✓ Commit all exercise outputs and findings to Git with structured README documentation
- ✓ Describe the G/VG distinction: description vs analysis-of-consequence-with-recommendation